🔧 Engine Teardown & Rebuild: Lab Sheet

Small Engines Overhaul · Unit 8 · Level 3 · goes with Engine project + Engine Assembly 1 + Ring End Gap Handout

Name: _____

Question: Is your engine's piston and valve condition within spec, and does the rebuilt engine run?

Materials

engine, tool set, flywheel puller, feeler gauges, torque wrench, lapping kit, parts tray, manual

⚠️ Safety

- Safety glasses and coveralls
- Plug wire disconnected; fluids drained with adult supervision
- Springs: eye protection, tools with adult supervision
- Start engine outdoors with an adult

Prediction

Predict which part will be most worn.

_____

_____

Procedure

1. Tear down in order with photos and labeled trays.
2. Measure ring end gap and side clearance.
3. Lap valves; set clearance.
4. Reassemble with timing aligned and bolts torqued.
5. Test fuel, spark, compression; run with adult.

Data

Check	Spec	Measured	Action
Top ring end gap			
Ring side clearance			
Intake valve clearance			
Exhaust valve clearance			
Head bolt torque			

Analysis (Claim, Evidence, Reasoning)

1. Claim: what was wrong with your engine and did the rebuild fix it? Evidence: your measurements and test. Reasoning.